

What argument is passed to the recursive function call? The node to traverse to, whose child nodes will be the next nodes to traverse.

How does this argument become closer to the base case? There are no cycles in a DAG, so following the descendant nodes will always eventually reach a leaf node.

Keep in mind that tree data structures that are especially deep will cause a stack overflow as the algorithm traverses the deeper nodes. This happens because each level deeper into the tree requires yet another function call, and too many function calls without returning cause stack overflows. However, it's unlikely for broad, well-balanced trees to be that deep. If every node in a 1,000 level deep tree has two children, the tree would have about 2^{1000} nodes. That's more atoms than there are in the universe, and it's unlikely your tree data structure is that big.

Trees have three kinds of tree traversal algorithms: preorder, postorder, and inorder. We'll discuss each of these in the next three sections.

Preorder Tree Traversal

Preorder tree traversal algorithms access a node's data before traversing its child nodes. Use a preorder traversal if your algorithm needs to access the data in parent nodes before the data in their child nodes. For example, preorder traversals are used when you are creating a copy of the tree data structure, as you need to create the parent nodes before child nodes in the duplicate tree.

The following *preorderTraversal.py* program has a `preorderTraverse()` function that traverses each child node first, before accessing the node's data to print it to the screen:

Python

```
root = {'data': 'A', 'children': [{'data': 'B', 'children':
[{'data': 'D', 'children': []}]}, {'data': 'C', 'children':
[{'data': 'E', 'children': [{'data': 'G', 'children': []},
{'data': 'H', 'children': []}]}, {'data': 'F', 'children': []}]}}

def preorderTraverse(node):
    print(node['data'], end=' ') # Access this node's data.
    ❶ if len(node['children']) > 0:
```